3.21 Modules over Local Rings (Nakayama's Lemma)

The main result of this section (3.21.7) is that every finitely-generated module over a local ring has a minimal spanning set. Recall in the vector space case a minimal spanning set is precisely a basis (2.3.8). Analogously we may also show that (in the local case) every minimal spanning set has the same order. The crucial result is Nakayama's Lemma, which we develop here.

Definition 3.21.1 (Jacobson Radical)
Let A be a commutative ring. Define the ***Jacobson Radical*** *to be the intersection of all maximal ideals*

$$\sqrt{0}^J := \bigcap_{\mathfrak{m} \lhd A} \mathfrak{m}$$

Proposition 3.21.2
The Jacobson Radical $\sqrt{0}^J$ is a proper ideal

Example 3.21.3
When $(A, \mathfrak{m}_A)$ is a local ring then $\sqrt{0}^J = \mathfrak{m}_A$.

Lemma 3.21.4 (Characterization of Jacobson Radical)
For an ideal $\mathfrak{a}$ and $\mathfrak{m}$ a maximal ideal

a) $\mathfrak{a} \not\subseteq \mathfrak{m} \iff \mathfrak{a} + \mathfrak{m} = A \iff (1 + \mathfrak{a}) \cap \mathfrak{m} \neq \varnothing$
b) $\mathfrak{a} \subseteq \sqrt{0}^J \iff 1 + \mathfrak{a} \subseteq A^\star$
c) $x \in \sqrt{0}^J \iff 1 + (x) \subseteq A^\star$

Proof. We prove each in turn.

a) Clearly $\mathfrak{a} \subseteq \mathfrak{m} \implies \mathfrak{a} + \mathfrak{m} = \mathfrak{m}$. Conversely $\mathfrak{a} \not\subseteq \mathfrak{m} \implies \mathfrak{a} + \mathfrak{m} = A$ by maximality. Suppose $\mathfrak{a} + \mathfrak{m} = A$ then $1 = a + m$ whence $(1 - a) \in \mathfrak{m}$. The converse is similar.

b) By a) $\mathfrak{a} \subseteq \sqrt{0}^J \implies (1 + \mathfrak{a}) \cap \mathfrak{m} = \varnothing$ for all maximal ideals $\mathfrak{m}$. By (3.3.15) this implies $(1 + \mathfrak{a}) \subseteq A^\star$.

Conversely if $(1 + \mathfrak{a}) \subseteq A^\star$ then $(1 + \mathfrak{a}) \cap \mathfrak{m} = \varnothing$ for any maximal ideal $\mathfrak{m}$ by (3.3.13). Again by a) $\mathfrak{a} \subseteq \mathfrak{m}$ as required.

c) This follows from b) and noting $x \in \sqrt{0}^J \iff (x) \subseteq \sqrt{0}^J$. ∎

Proposition 3.21.5 (Nakayama's Lemma)
Let M be a finitely generated A-module and $\mathfrak{a} \lhd A$ an ideal. Then the following holds

a) If $M = \mathfrak{a}M$ then there exists $a \in \mathfrak{a}$ such that $m = am$ for all $m \in M$

Suppose in addition that $\mathfrak{a} \subseteq \sqrt{0}^J$ (e.g. if A is local and $\mathfrak{a}$ is proper) then

b) $M = \mathfrak{a}M \implies M = 0$
c) $N \leq M$ *and* $M = N + \mathfrak{a}M \implies M = N$.

Proof. We prove each in turn

a) Apply (3.18.4) with $\phi := \mathbf{1}_M$ to find a monic polynomial $P(X) \in A[X]$ with non-leading coefficients in $\mathfrak{a}$ such that $P(\phi)(m) = 0$ for all $m \in M$. Then we see that $(1 + a_{n-1} + ...a_0)m = 0$ for all $m \in M$ whence $a := -(a_{n-1} + ... + a_0)$ is the required element.

More directly, suppose $m_1, ..., m_n$ is a generating set for M. By (3.18.3) (and $M = \mathfrak{a}M$) there is a matrix E with coefficients in $\mathfrak{a}$ such that

$$(I_n - E)\boldsymbol{m} = 0$$

where $\boldsymbol{m}$ is the column vector consisting of $m_1, ..., m_n$. By (3.5.23) we see $\det(I_n - E)m_i = 0$ for all $i = 1...n$. It's enough to show $a := \det(I_n - E) \in 1 + \mathfrak{a}$. Observe

$$\det(I_n - E) = \prod_i (1 - E_{ii}) + \sum_{\sigma \neq \mathrm{id}} \epsilon(\sigma) \prod_j E_{j\sigma(j)}$$

The second term lies in $\mathfrak{a}$ and

$$\prod_i (1 - E_{ii}) = 1 - \sum_{i=1}^n E_{ii} \prod_{j>i} (1 - E_{jj}) \in 1 + \mathfrak{a}$$

b) Consider any $m \in M$. By a) we have $(1-a)m = 0$ for some $a \in \mathfrak{a}$, and by (3.21.4) $(1-a)$ is invertible, whence $m = 0$ as required.

c) Observe $\mathfrak{a}(M/N) \overset{(3.3.106)}{=} (N + \mathfrak{a}M)/N = M/N$ whence $M/N = 0$ by b). Therefore $N = M$ as required.

We may show b) more directly. Suppose $M \neq 0$, and let

$$\{m_1, ..., m_n\}$$

be a non-zero generating set for M of minimal size. Then by (3.18.3)

$$m_1 = \sum_j a_j m_j \quad a_j \in \mathfrak{a}$$

whence

$$(1-a_1)m_1 = \sum_{j \geq 2} a_j m_j$$

As $a_1 \in \sqrt{0}^J$ we have $1 - a_1 \in A^\star$ by (3.21.4). Then

$$\{m_2, ..., m_n\}$$

is a smaller generating set, a contradiction. Therefore $M = 0$.

a) may be deduced from b) as follows. Observe $S \coloneqq 1 + \mathfrak{a}$ is a multiplicatively closed subset, so we may consider $S^{-1}M$ as an $S^{-1}A$-module. It's easy to verify that $1 + S^{-1}\mathfrak{a} \subseteq (S^{-1}A)^\star$ so by (3.21.4) $S^{-1}\mathfrak{a} \subseteq J(S^{-1}A)$. Clearly $\mathfrak{a}M = M \implies (S^{-1}\mathfrak{a})S^{-1}M = S^{-1}M$ so by the weaker form $S^{-1}M = 0$. By (3.6.16) there exists $s \in S$ such that $sM = 0$, which is the required result as $s = 1 + a$ for some $a \in \mathfrak{a}$. ∎

Recall (3.3.107) in the case of a local ring $(A, \mathfrak{m})$ that $\widetilde{M} \coloneqq M/\mathfrak{m}M$ is a vector space over $k \coloneqq A/\mathfrak{m}$. We may use Nakayama's Lemma to exhibit a correspondence between minimal spanning sets of M and bases of $M/\mathfrak{m}M$ as a k-vector space. First we prove a simpler form

Lemma 3.21.6
Let $(A, \mathfrak{m})$ be a local ring with residue field $k = A/\mathfrak{m}$, M a finite A-module and $S \subset M$ a subset. Then

$$S \text{ spans } M \iff \tilde{S} \text{ spans } \widetilde{M}$$

where $\tilde{}$ denotes reduction modulo $\mathfrak{m}M$.

Proof. One direction is obvious. Conversely suppose $\tilde{S}$ spans $\widetilde{M}$. Define $N \coloneqq \langle S \rangle$, then this means precisely that $N + \mathfrak{m}M = M$, so $N = M$ by (3.21.5) c) as required. ∎

Recall from (2.1.57) that every subset of T of $M/\mathfrak{m}M$ may be written in the form $\tilde{S}$ for $S \subset M$ and $\tilde{}$ injective on S.

Proposition 3.21.7 (Structure theorem for modules over a local ring)
Let $(A, \mathfrak{m})$ be a local ring with residue field $k = A/\mathfrak{m}$, M a finite A-module. Then

a) $\widetilde{M} \coloneqq M/\mathfrak{m}M$ is a finite-dimensional k-module (of dimension n say)
b) If $\tilde{S}$ is a k-basis for $\widetilde{M}$ and $\tilde{}|_S$ is injective, then S is a minimal spanning set for M and $\#S = \#\tilde{S} = n$
c) If S is a minimal spanning set then $\tilde{S}$ is a basis for $\widetilde{M}$ (and $\tilde{}|_S$ is injective so $\#S = \#\tilde{S} = n$).

Proof. a) By (3.3.107) $\widetilde{M}$ is a k-module, and it's clearly finite.

b) By (3.21.6) S spans M. Suppose $S' \subset S$ spans M, then by the same result $\widetilde{S'}$ spans $\widetilde{M}$. Recall (2.3.8) that a vector space basis is precisely a minimal spanning set, so $\widetilde{S'} = \tilde{S}$. As $\tilde{}$ is injective this means $S' = S$. Therefore S is a minimal spanning set.

c) Let S be a minimal spanning set. Then by (3.21.6) $\tilde{S}$ spans $\widetilde{M}$. Suppose $\tilde{T} \subset \tilde{S}$ also spans $\widetilde{M}$. By (3.21.6) T spans M, and by hypothesis $T = S$. Therefore $\tilde{T} = \tilde{S}$. As $\tilde{T}$ was arbitrary we see that $\tilde{S}$ is a minimal spanning set for $\widetilde{M}$, which by (2.3.8) is a basis.

Finally suppose $\tilde{}$ is not injective on S, that is $\tilde{s_1} = \tilde{s_2}$. Then $S' \coloneqq S \setminus \{s_1\}$ satisfies $\widetilde{S'} = \tilde{S}$. Therefore by the Lemma S' spans M, contradicting minimality. ∎